American Academy of Political and Social Science

Three Phases in the Evolution of Risk Communication Practice
Author(s): William Leiss
Source: *Annals of the American Academy of Political and Social Science*, Vol. 545, Challenges in Risk Assessment and Risk Management (May, 1996), pp. 85-94
Published by: Sage Publications, Inc. in association with the American Academy of Political and Social Science
Stable URL: http://www.jstor.org/stable/1047895
Accessed: 12/09/2008 09:24

Your use of the JSTOR archive indicates your acceptance of JSTOR's Terms and Conditions of Use, available at http://www.jstor.org/page/info/about/policies/terms.jsp. JSTOR's Terms and Conditions of Use provides, in part, that unless you have obtained prior permission, you may not download an entire issue of a journal or multiple copies of articles, and you may use content in the JSTOR archive only for your personal, non-commercial use.

Please contact the publisher regarding any further use of this work. Publisher contact information may be obtained at http://www.jstor.org/action/showPublisher?publisherCode=sage.

Each copy of any part of a JSTOR transmission must contain the same copyright notice that appears on the screen or printed page of such transmission.

JSTOR is a not-for-profit organization founded in 1995 to build trusted digital archives for scholarship. We work with the scholarly community to preserve their work and the materials they rely upon, and to build a common research platform that promotes the discovery and use of these resources. For more information about JSTOR, please contact support@jstor.org.

Sage Publications, Inc. and *American Academy of Political and Social Science* are collaborating with JSTOR to digitize, preserve and extend access to *Annals of the American Academy of Political and Social Science.*

http://www.jstor.org

ANNALS, *AAPSS*, **545**, May 1996

Three Phases in the Evolution of Risk Communication Practice

By WILLIAM LEISS

ABSTRACT: Effective communication between interested parties is widely held to be a vital element in health and environmental risk management decision making. There have been three phases in the evolution of risk communication during the last twenty years. Phase I emphasized risk: in a modern industrial economy, we must have the capacity to manage risks at a very exacting level of detail. Phase II stresses communication: statements about risk situations are best regarded as acts of persuasive communication, that is, as messages intended to persuade a listener of the correctness of a point of view. Now, in Phase III, public and private sector institutions increasingly are recognizing their responsibility to deal adequately with both dimensions and to carry out sound risk communication as a matter of good business practice.

William Leiss is a fellow of the Royal Society of Canada and holds the Eco-Research Chair in Environmental Policy at Queen's University, Kingston, Ontario. He chairs the Research Policy Committee of the Canadian Global Change Program. He has held academic appointments in political science, sociology, environmental studies, and communication. He is author, collaborator, or editor for nine books and numerous articles. He works extensively with industry and government in the area of risk communication, risk management, public consultation, and consensus building.

RISK communication may be defined as the flow of information and risk evaluations back and forth between academic experts, regulatory practitioners, interest groups, and the general public. The sharp disagreements that can occur between members of these constituencies over the best ways to assess or manage risks sometimes are based on disagreements over principles or approaches, sometimes on differences in the information base available to various parties, and sometimes on a failure to consider carefully each other's position.

In such situations, the risk communication process itself often becomes an explicit focus of controversy. Charges of media bias or sensationalism, of distorted or selective use of information by advocates, of hidden agendas or irrational standpoints, and of the inability or unwillingness of regulatory agencies to communicate vital information in a language the public can understand are common. Such charges are traded frequently at public hearings, judicial proceedings, and conferences, expressing the general and pervasive sense of mistrust felt by many participants toward others. Of course, there are also genuine differences in principle, outlook, and values in the citizenry; disagreements will persist even with the most complete and dispassionate knowledge of others' views. Perhaps the most contentious area of all is that of risk-benefit trade-offs, especially where different types (or distributions) of risk are at stake, or where there is no consensus on acceptable risk (thus preventing trade-offs).

Risk communication research, which seeks to clarify our understanding of the processes just described briefly, is the newest of the four risk subfields; the phrase itself appears to have been coined during 1984.[1] It arose out of the problems being investigated in the risk perception area, which since its inception had concentrated on the disparities between risks as assessed by experts, on the one hand, and as understood by the general public, on the other. Risk perception studies have been concerned with explaining those disparities. The interest in risk communication, however, has from the beginning had a practical intent: given that these disparities exist, are deeply entrenched in human awareness, and form the basis of strongly held attitudes and behavior, how can we improve the quality of the dialogue about risk across the gap that separates experts from the general public? Second, how can we apply this improved dialogue to achieving a higher degree of social consensus on the inherently controversial aspects of managing environmental and health risks?

In seeking to answer these questions, risk communication researchers have married their knowledge about risk assessment and management issues with the approaches used in the field of modern communication the-

1. This is the year of the earliest uses of the phrase "risk communication," according to the references listed in Bernd Rohrmann, Peter M. Wiedemann, and Helmut U. Stegelmann, eds., *Risk Communication: An Interdisciplinary Bibliography*, 4th ed. (Jülich, Germany: Research Center Jülich GmbH, 1990), pp. 26, 56, 111.

ory and practice. Statements about risk by various parties are treated as messages intended to persuade others to believe or do something; like all such messages circulating among persons, their effectiveness as acts of persuasive communication can be evaluated according to well-established criteria: whether they gain attention, are understood, are believed, are acted on, and so forth. This paradigm of communication research has become very well established since 1945 and has an enormous published literature to support it.[2] Risk communication research has been able to draw on this resource and adapt its findings to the particular concerns of the risk studies area, and as a result it has made substantial progress in a relatively short time.

THE THREE PHASES

There are three phases in the evolution of risk communication, occurring over the past 15 years, and each of the later stages has emerged in response to the earlier ones. The earlier ones do not become irrelevant; rather, they are incorporated into the later phases, for each has contributed something of lasting value to the present. Phase I (about 1975-84) stressed the quantitative expressions of risk estimates and argued that priorities for regulatory actions and public concerns should be established on the basis of comparative risk estimates.[3] Phase II (about 1985-94) stressed the characteristics of successful communications: source credibility, message clarity, effective use of channels, and, above all, a focus on the needs and perceived reality of the audiences.[4] Around 1995, we entered a new phase, which will be described briefly later.

One of the leading authorities in the risk studies field, Baruch Fischhoff of Carnegie-Mellon University, recently presented a somewhat different account of the developmental stages in risk communication. He used colloquial expressions to identify seven such stages:

1. "All we have to do is get the numbers right."
2. "All we have to do is tell them the numbers."
3. "All we have to do is explain what we mean by the numbers."
4. "All we have to do is show them that they've accepted similar risks in the past."
5. "All we have to do is to show them that it's a good deal for them."
6. "All we have to do is treat them nice."
7. "All we have to do is make them partners."[5]

2. William Leiss, "On the Vitality of Our Discipline: New Applications of Communications Theory," *Canadian Journal of Communication*, 16:291-305 (1991).

3. The wide purview now enjoyed by the risk approach stems from a few sources, including W. W. Lowrance, *Of Acceptable Risk* (Los Altos, CA: Wm. Kaufmann, 1976); William D. Rowe, *An Anatomy of Risk* (New York: John Wiley, 1977). Any such separation of a dynamic process into phases and dates is somewhat arbitrary; the activities in each phase overlap.

4. The seminal work is Vincent T. Covello, Detlof von Winterfeldt, and Paul Slovic, *Risk Communication: Background Report for the National Conference on Risk Communication* (Washington, DC: Conservation Foundation, 1986).

5. Baruch Fischhoff, "Risk Perception and Communication Unplugged: Twenty Years of Process," *Risk Analysis*, 15:137-45 (1995).

The first two correspond roughly to my Phase I, the next four to my Phase II, and the last one to the current phase. Any such typology is arbitrary. The one I have devised highlights the radical nature of the transition from Phase I to Phase II, and in my view, the field of risk communication as we know it today was formed by this wrenching transition.

Phase I (1975-84)

The enduring strength of what was accomplished in Phase I is captured in the following statement: in order to function sensibly in a world of expanding opportunity, we must have the capacity to assess and manage risks at a very exacting level of detail; the scientific approach to risk management offers us an imperfect but indispensable tool for doing so. Although risk is conventionally understood as "exposure to the chance of loss," we derive enormous benefits from judicious risk-taking behavior, so long as we are clever enough to know where to draw the line. For example, industrial chemicals are the basis of most consumer goods today, but all of them are also dangerous in certain doses; we have to know what the doses are that are likely to produce adverse effects on human health and the environment, and we must have institutional mechanisms in place to ensure that we do not exceed those doses.

Some serious weaknesses emerged in this phase, the worst of which could be labeled the "arrogance of technical expertise." Faced with public opposition to the results of risk-based decision making, many experts responded with open contempt toward the public perception of risk.[6] For them, perceived risk is correlated with false understanding and is further contrasted with real risk, which is allegedly an objective—that is, "true"—account of reality. Fortunately, one encounters this invidious distinction less and less now, since there is a greater appreciation of the errors in judgment that experts are prone to making.[7] Partly as a result of the arrogance of expertise, there exists on the part of the public a profound distrust of experts and the institutions they represent, which weakens the force of the quite sensible contributions that technical experts can make to the public discourse on risk taking. Another weakness is that critical data gaps and ever changing scientific research results are common in all significant risk management areas; the uncertainties introduced thereby produce legitimate concerns when yes/no decisions must be made.

The underlying message of permanent value in Phase I is that, for individuals as well as societies, managing opportunities and dangers on the basis of comparative risk information is an inescapable duty of intelligent life. However, this message could not be communicated effec-

6. See, for example, Ernest Siddall and Carl R. Bennett, "A People-Centered Concept of Society-Wide Risk Management," in *Environmental Health Risks: Assessment and Management*, ed. R. Stephen McColl (Waterloo, ON: University of Waterloo Press, 1987), p. 272.

7. A good summary of the errors in judgment to which experts are prone is National Research Council, *Improving Risk Communication* (Washington, DC: National Academy Press, 1989), pp. 44-47.

tively to a wide range of public audiences, partly because its authors were often so openly contemptuous of the fundamental beliefs about risk taking that were held by the very audiences whom they were addressing.

Phase II (1985-94)

The radical break that defines the transition from Phase I to Phase II was the realization that statements about risk situations ought to be regarded as acts of persuasive communication, that is, as messages intended to persuade a listener of the correctness of a point of view.[8] Guidance for this new approach was found in the history of twentieth-century marketing communications, which had demonstrated—first in commercial advertising, then more broadly—the effectiveness of a strategy that takes into account two key factors: the characteristics of the audience itself, and the intrinsic legitimacy of the audience's perception of the situation. The coinage of good communication is trust in the message source ("Will you believe me when I tell you something?"), and this is the underpinning for credibility, which is a perception of the intrinsic honesty of the whole process.

The great strength of this new approach was that the formulae of good communication practices adapted from modern marketing had been tested and refined in minute detail over a long period and, for some purposes at least, were known to be highly successful. But there proved to be severe difficulties in adapting this marketing communication paradigm to risk issues. Slovic and MacGregor have diagnosed the main problem well:

> Although attention to communication can prevent blunders that exacerbate conflict, there is rather little evidence that risk communication has made any significant contribution to reducing the gap between technical risk assessments and public perceptions or to facilitating decisions about nuclear waste or any other major sources of risk conflict. The limited effectiveness of risk communication efforts can be attributed to the lack of trust.[9]

The paradigm of persuasion in the marketing communication approach had identified a broad range of techniques for enhancing trust and credibility for messages. The early studies on propaganda already had recognized, however, that too strong a focus on the persuasive techniques themselves—especially those that seek to manipulate audiences' emotions—is potentially dangerous, for it could result in any rational content in the message being subordinated or even dissolved by those excessively clever techniques. In the more prosaic world of risk issues, therefore, emotive techniques of effective persuasive communication—that is, techniques for convincing that audience that a particular person is a

8. Another way of putting this point is to say that "risk is a construct," that is, an understanding of a risk-type situation that is related to the situation of each participant. See Bayerische Rück, ed., *Risk Is a Construct* (Munich: Knesebeck, 1993). This transition represents a complete change of emphasis within the components of the phrase "risk communication": in Phase I, the emphasis is on the adjective; in Phase II, on the noun.

9. Paul Slovic and Donald J. MacGregor, "The Social Context of Risk Communication" (Paper, Decision Research, Eugene, OR, 5 May 1994), p. 17.

credible spokesperson on risk issues—could take precedence over the informational content of the risk message itself.[10]

The underlying message of permanent value in Phase II may be stated as follows: there is an obligation on the part of major institutional actors in society to communicate effectively about risks, not by simply touting the superiority of their own technical risk assessments but, rather, by making an honest effort to understand the bases of public risk perceptions and by experimenting with ways of constructing a reasoned dialogue around different stakeholder assessments of risk situations.[11] The residual weakness here is that trust is often far too low for these experiments to succeed.

Phase III (current)

Phase III starts with the recognition that that lack of trust is pervasive in risk issues and that, because of this, risk communication practice must move away from a focus on purely instrumental techniques of persuasive communication. Phase III is characterized by an emphasis on social context, that is, on the social interrelations between the players in the game of risk management.[12] It is based on the presumption that, despite the controversial nature of many risk management issues, there are forces at work also that favor consensus building, meaningful stakeholder interaction, and acceptance of reasonable government regulatory frameworks. Should those forces turn out to be relatively weak, both public sector fiscal constraint and the current delegitimization of government may leave the field of risk management exposed to wide-open confrontation between stakeholder interests.

As noted earlier, Phase II remained incomplete because the key ingredient of successful persuasive communication—trust—cannot be manufactured by the use of techniques alone, no matter how artful the practitioners are. A working hypothesis is that trust in institutional risk actors (governments and industry) can accumulate, slowly, through the commitments by those institutions—as demonstrated by deeds, not words—to carry out responsible risk communication and, furthermore, to do so consistently, as a matter of daily practice over the long term, not just in response to crisis events. At the moment, there is no code of good practice in this area that might provide some benchmarks for determining what is and what is not responsible risk communication, although I suspect that events during Phase III will lead in that direction. For now, we will have to make do with case study examples, two of which follow in the concluding section.

Thus the underlying message of permanent value in Phase III may be stated as follows: a demonstrated

10. In this period, institutional risk managers (government and industry) were often told that up to 75 percent of message content—as received by audiences—was based on the nonverbal dimensions of the message delivery format itself: body posture, hand gestures, style of dress, facial expression, and so forth.

11. See William Leiss, " 'Down and Dirty': The Use and Abuse of Public Trust in Risk Communication," *Risk Analysis* 15:685-92 (1995).

12. The best guide is Slovic and MacGregor, "Social Context of Risk Communication."

commitment to responsible risk communication by major organizational actors can put pressure on all players in risk management to act responsibly.

PHASE III CASE STUDIES

A typical Phase III case involves a company that has a sensitive issue, as well as one or more documents in highly technical language, that must be discussed with nonexpert stakeholders (employees, community associations).[13] The two cases that will be identified here occurred in Canada during 1995. Responsible risk communication practice suggests that it is unacceptable to simply distribute the relevant document or documents to the public with a cover note saying, "Here it is; you figure out what it's supposed to mean to you." The risk communication challenge takes this form: the documents provide the information, but what is the message?

Case study 1: Dow Chemical Canada Inc., report on dioxin emissions from vinyl plant stacks

Dow Chemical Canada voluntarily took responsibility to develop, in cooperation with government agencies, a protocol to measure dioxin emissions from the stacks at its vinyl plant in Fort Saskatchewan, Alberta. Then the company commissioned an independent, university-based expert group to do an exposure assessment for areas within a certain radius of the plant. The expert group's report contains various figures for "incremental" additions to "background" dioxin levels, with the estimated human intakes expressed as picograms per kilogram of body weight per day (the relevant Canadian guideline is 10 pg/kg/d).[14]

Prior to communicating publicly about this exposure assessment, the company undertook a series of internal meetings to discuss the most effective way of communicating this information to the community. All of the following statements, as well as many variants that were considered, had to pass a "threshold criterion," namely, each one had to be believed to be a factually truthful statement by everyone who participated in these exercises. With this criterion in place, and recognizing the considerable challenge in the technical description of exposure assessment contained in the report, the objective of the company's "key messages" was to convey its own understanding of the "bottom-line" conclusions that it contained. The first two statements in the following list were selected as the bases for framing the communication message.

"The [Institute for Risk Research] Report results show that

— "dioxin emissions from our vinyl plant do not add significantly to the existing 'background exposure' level for dioxins";

13. The only published case study of this type of which I am aware is Caron Chess et al., "The Organizational Links Between Risk Communication and Risk Management: The Case of Sybron Chemicals, Inc.," *Risk Analysis*, 12:431-38 (1992).

14. John Hicks and Stephen McColl, "Final Report: Exposure Assessment of Airborne Dioxins and Furans Emitted from the EDC/VCM Facility at the Dow Chemical Canada Fort Saskatchewan Site" (Report, Institute for Risk Research, University of Waterloo, ON, 31 Mar. 1995).

— "all together, dioxin levels at and nearby the plant site are well within the current Health Canada guidelines";
— "the incremental contribution to background levels from the vinyl plant is exceedingly small";
— "the incremental contribution . . . is insignificant";
— "the incremental contribution . . . is very small, that is, less than ___% of background levels and ___% of the Health Canada guideline";[15]
— "in assessing total human exposure to dioxins, an insignificant portion is attributable to the emissions from Dow sources";
— "current background levels of dioxins in Fort Saskatchewan, and virtually everywhere in North America, are known to be about 3.2 pg/kg/day. The vinyl plant stacks at the Dow site add about ___% additional exposure to the background level. All together, these numbers are still well below the current Health Canada guideline of 10 pg/kg/day."

It is important to note as well that the formulation of these key messages, considered as an exercise in responsible risk communication, occurred in the context of a broader risk communication strategy. This broader strategy included, first and foremost, the decision to commission an independent group to undertake the study, which, of course, also entailed making it public in exactly the form in which it was written. Second, the company undertook to bring the study's principal author to a meeting of its Citizen Advisory Panel in Fort Saskatchewan, where a face-to-face discussion of the report's findings could take place.

Case study 2: CXY Chemicals (North Vancouver plant), worst-case scenario at a chlorine plant located in an urban area

A quantitative risk assessment (QRA) of the public safety risk associated with the plant was undertaken by an independent consulting firm. The QRA report concluded that the worst case was a possible breach of all of the plant's pressurized storage tanks, containing a maximum total of 1500 tonnes of chlorine, as a result of a severe earthquake (a 1-in-475-year event); a second scenario assumed the possibility of a much reduced chlorine inventory. Air movement (wind speed and direction) at the site was a prime factor in the "risk contours" that were developed in the QRA; the contours are expressed as elongated circles showing varying probabilities of fatalities as a result of the hypothesized event.[16] The results showed that, under conditions of operation prevailing at the time of the study, the plant would not meet criteria established by the Major Industrial Accidents Council of Canada (MIACC)

15. The actual percentages depend on which section of the Institute for Risk Research report is being referred to.

16. Ertugrul Alp et al., "CanadianOxy North Vancouver Plant: Quantitative Assessment of Safety Risks" (Report, Bovar–Concord Environmental, Downsview, Ontario, Aug. 1994). CanadianOxy changed its name to CXY Chemicals in 1995.

for acceptable risk for land uses in urban areas.[17]

Prior to communicating publicly about the QRA report, the company undertook a series of internal meetings, evaluating a variety of options, in order to choose how to respond to its findings and to formulate the company's "key messages" at the time when the QRA report would be publicly released. A particularly difficult challenge was in choosing a method for the communication process about the QRA methodology itself, with the need to explain how such a scenario is constructed, including the use of worst-case assumptions for every relevant risk parameter, without regard to probability of occurrence.[18] However successful this communication effort might hope to be, though, the bottom line was that the current plant inventory, as a result of the study findings, would be perceived to be beyond a reasonable contemporary standard for acceptable risk. The company could only respond to this challenge by first making a series of management decisions to reduce the risks associated with plant operations.

The QRA report contained a number of recommendations for immediate risk reduction, in areas peripheral to the main plant operation itself; these were all implemented. The most significant recommendation, also implemented, was to change the plant's inventory management, so that no more than 300 tonnes of chlorine—one-fifth of the level assumed for the worst-case scenario—would be stored at any time. When the risk contours were recalculated using this operational directive, the plant operations fell within the MIACC criteria for acceptable risk.

The risk communication process associated with the QRA study then came to be viewed as an integral part of the risk management decision making that had led to significant risk reduction. The company had always been committed to commissioning an independent group to undertake the study, which, of course, also entailed making the study public. Second, the company has undertaken to hold face-to-face discussions on the original QRA report's findings with any individuals or local groups who request such a meeting. The company's key messages focus on what it regards as that part of the entire story that is of immediate and practical relevance to its employees and to residents of surrounding communities, namely, the company's ongoing commitment to risk management, which had resulted in significant risk reduction. Those key messages are that

— "CXY Chemicals has safely manufactured, stored and transported chlorine and related products from the North Vancouver plant for 37 years."
— "CXY Chemicals, as an industrial chemical manufacturer, is proud of its safety record and continues to take steps to reduce exposure to risk."

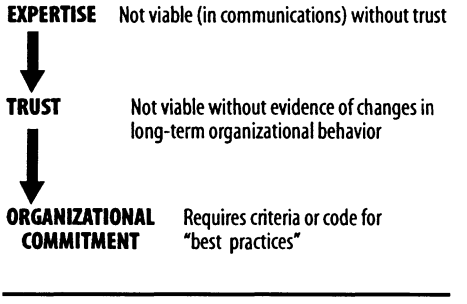
17. MIACC criteria suggest a buffer zone around a risk source and various types of restricted land uses up to the point where the annual fatality risk is no greater than one in a million.

18. Cf. Wil Leplowski, "Chemical Companies Make Public Worst Case Accident Scenarios," *Chemical & Engineering News*, 20 June 1994, pp. 22-26.

CONCLUSION AND EXPECTED FURTHER DEVELOPMENTS

The perspective taken in this article is that the specific features of each phase in the evolution of risk communication strategies set the challenges for the period to follow. The weaknesses in each phase drive this evolution further, and in this process we strive to preserve the strengths of each. The three-phase evolutionary development is summarized in Figure 1.

FIGURE 1
OVERVIEW OF THE THREE PHASES



A good theoretical framework for Phase III may be found by extending the "strategic environmental audit" and "environmental responsibility" approach.[19] This could be operationalized by the formulation of a "code of good risk communication practice," and compliance with the code could be verified through a "risk communication audit" designed to meet the test of public credibility. Some of the much-needed foundations of trust might be laid in this manner.

19. Grant Ledgerwood et al., *Implementing an Environmental Audit: How to Gain Competitive Advantage Using Quality and Environmental Responsibility* (Homewood, IL: Irwin, 1994).